# Ibnu Cipta Ramadhan

Edge AI | Computer Vision | IoT Engineer

ibnucr@email.com ibdhan.github.io ibdhan linkedin
Surabaya, Indonesia (Available for Remote Collaboration)

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## Professional Summary

Embedded AI, IoT, and Computer Vision Engineer with over five years of experience developing intelligent edge systems, from embedded hardware and sensor integration to real-time AI deployment. Experienced in building production IoT devices, edge-to-cloud solutions, robotics perception systems, and computer vision applications using NVIDIA Jetson, ROS 2, and PyTorch. Comfortable working across the full engineering stack, from electronics and firmware to AI optimization, device integration, and scalable IoT system development.

## Skills

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| **AI / CV** | 2D and 3D Object Detection, YOLO, OpenCV, PyTorch, TensorRT |
| **Edge / Embedded** | NVIDIA Jetson (Orin, Xavier), CUDA, ROS 2, LiDAR |
| **IoT** | GPS/Fleet Tracking, MQTT, LoRa, Telemetry, Redash, Metabase |
| **Programming** | Python, C, C++, SQL, Bash, LaTeX |
| **Development** | Docker, Git, Agile, Scrum, AWS |

## Work Experience

**University Lecturer** *2024.08 – Present*
*Politeknik Elektronika Negeri Surabaya (PENS)* Surabaya, Indonesia

- Conduct research in edge AI systems in computer vision, robotics, and embedded intelligence.
- Supervised undergraduate engineering projects involving AI, robotics, and IoT deployment.
- Designed laboratory modules covering sensor & actuators, instrumentation, and digital signal processing.

**Senior Hardware & Internet of Things (IoT) Engineer** *2024.02 – 2024.07*
*McEasy – PT. Otto Menara Globalindo* Surabaya & Jakarta, Indonesia

- Designed end-to-end IoT solutions for logistics and fleet management applications.
- Worked directly with enterprise customers to translate operational requirements into deployable hardware and software solutions.
- Collaborated with cross-functional software teams to integrate embedded devices into cloud platforms.
- Supported field deployment, debugging, and hardware validation.

**Hardware & Internet of Things (IoT) Engineer** *2023.02 – 2024.01*
*McEasy – PT. Otto Menara Globalindo* Surabaya & Jakarta, Indonesia

- Integrated GPS tracking devices and telematics sensors with fleet monitoring platforms.
- Designed and executed test cases for GPS trackers, sensors, and embedded hardware before production deployment.
- Analyzed telemetry data to investigate product issues, evaluate device performance, and support data-driven feature development.
- Developed hardware and sensor-related features based on customer and product requirements.

**Research Assistant - Collision Avoidance System Development For Mining** *2023.02 – 2023.05*
*STEI – ITB* Bandung, Indonesia

- Developed perception and sensing modules for collision avoidance systems in mining vehicles.
- Integrated LiDAR, camera, and sensor data for environment perception and obstacle detection.
- Evaluated system performance through simulation and field testing.

### Research Assistant - Autonomous Tram Development Using AI
*2021.01 – 2023.04*
*STEI – ITB* — Bandung, Indonesia

- Developed AI-based perception systems for autonomous tram navigation using LiDAR data.
- Implemented and evaluated 3D object detection algorithms on NVIDIA DRIVE AGX platforms.
- Collected, processed, and analyzed real-world datasets for model development and validation.

### Internet of Things (IoT) Engineer
*2018.02 – 2020.08*
*Mceasy – PT. Otto Menara Globalindo* — Surabaya & Jakarta, Indonesia

- Designed embedded hardware for GPS tracking devices.
- Integrated GNSS, cellular communication, accelerometers, and digital I/O.
- Performed hardware bring-up, firmware validation, and manufacturing support.
- Worked closely with firmware and backend teams for production deployment.

### Internet of Things (IoT) Engineer (Part-time)
*2017.04 – 2018.02*
*Mceasy – PT. Otto Menara Globalindo* — Surabaya & Jakarta, Indonesia

- Designed embedded hardware for GPS tracking devices.
- Integrated GNSS, cellular communication, accelerometers, and digital I/O.
- Performed hardware bring-up, firmware validation, and manufacturing support.
- Worked closely with firmware and backend teams for production deployment.

### Internship
*2015.08 – 2015.09*
*PT. Pertamina Hulu Energi – West Madura Offshore* — Gresik, Indonesia

- Assisted engineers in maintenance and inspection activities for offshore production facilities.
- Supported data collection and technical documentation for engineering operations.

## Projects

### LiDAR-Based 3D Perception for Autonomous Tram
*2021 – 2023*
*Tech: PyTorch • CenterPoint • LiDAR • NVIDIA DRIVE AGX Pegasus*

- Developed a LiDAR-based 3D perception pipeline for autonomous tram navigation.
- Collected, processed, and annotated large-scale point cloud datasets for model training and evaluation.
- Deployed on NVIDIA DRIVE AGX Pegasus; achieved 20 FPS on real-world point cloud data
- Demo: https://www.youtube.com/watch?v=Dft3OKFTdvs

### GPS Fleet Tracking Platform
*2023 – 2024*
*Tech: GPS • LTE • IoT • Embedded Systems • Sensor Integration*

- Integrated GPS tracking devices and telematics sensors with cloud-based fleet management platforms.
- Designed validation procedures and test cases for GPS trackers and embedded hardware.
- Developed hardware and sensor-related features to improve system functionality and reliability.

### Edge AI Deployment on NVIDIA Jetson
*2025 – 2026*
*Tech: Jetson Xavier NX • Jetson Orin Nano • TensorRT • CUDA • Docker*

- Deployed and optimized deep learning models on NVIDIA Jetson edge devices.
- Benchmarked inference performance using TensorRT on multiple Jetson platforms.
- Developed containerized deployment workflows using Docker for reproducible development.
- Repo: https://github.com/ibdhan/ros2-yolo

### Real-Time Indonesian Traffic Sign Detection
*2025*
*Tech: YOLO • PyTorch • OpenCV • Computer Vision*

- Developed an object detection model for Indonesian traffic sign recognition using YOLO.
- Evaluated transfer learning and training-from-scratch approaches on a custom dataset.
- Optimized model performance for real-time edge deployment.

### IoT Telematics Monitoring Platform — *2023 – 2024*

*Tech: Python • PostgreSQL • Redash • Metabase • IoT*

- Developed data processing pipelines for GPS and telematics sensor data.
- Built monitoring dashboards and analytical tools for fleet performance and device health.
- Analyzed telemetry data to support feature development and operational decision-making.

## Education

### Master of Science (M.Sc.) – Electrical Engineering — *2020 – 2023*

*Institut Teknologi Bandung (ITB)* — Bandung, Indonesia

- **Thesis:** LiDAR-Based 3-Dimensional Object Detection for Autonomous Tram Perception System

### Bachelor of Applied Science (B.ASc.) – Electronics Engineering — *2016 – 2018*

*Politeknik Elektronika Negeri Surabaya (PENS)* — Surabaya, Indonesia

- **Thesis:** Binaural Sound Source Localization System Based on Fast Fourier Transform Using Interaural Phase Difference as Cue

### Diploma in Electronics Engineering — *2013 – 2016*

*Politeknik Elektronika Negeri Surabaya (PENS)* — Surabaya, Indonesia

- **Thesis:** Gas Monitoring System Oxygen, Carbon Monoxide, and Carbon Dioxide, with Integration on Wearable Device

## Languages

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| **English** | Professional working proficiency |
| **Bahasa Indonesia** | Native proficiency |